

Siddharth Parakh

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🔗 dufka54.github.io

SUMMARY

Analytical Computer & Communication Engineering undergraduate specializing in Machine Learning and Deep Learning architectures. Proficient in PyTorch, Python, and SQL for developing data-driven predictive systems, custom neural network components, and automated data pipelines.

EDUCATION

Manipal University Jaipur 08/2023 – 05/2027 (Expected) | Jaipur, Rajasthan
Bachelor of Technology - Computer and Communication Engineering
Overall CGPA: 8.43

Masai School, Minor in Artificial Intelligence and Data Science, IIT Mandi 06/2025 – 05/2026 | Distance Learning
Overall CGPA: 7.75

SKILLS

Languages (Python, SQL, C++, HTML, CSS) | **Libraries** (Pandas, NumPy, Matplotlib, Scikit-Learn, PyTorch) | **Tools** (Jupyter Notebook, MySQL, Google Colab, Streamlit, VS Code)

PROJECTS

MNIST Translation Invariance Analysis: FNN vs. CNN Performance - PyTorch | Matplotlib

- Designed and trained a Feedforward Neural Network (FNN) and a Convolutional Neural Network (CNN) from scratch using **PyTorch** to evaluate structural variations in digit classification on the MNIST dataset.
- Leveraged convolutional localized feature extraction and spatial weight-sharing to design a progressive filter model (1 -> 16 -> 32 -> 64) that achieved a **99.31% peak validation accuracy using only 46,570 trainable parameters** — delivering a nearly **10x computational footprint reduction** over the rigid 476,490-parameter FNN.
- Developed a custom data pipeline using **Pandas** and **Torchvision** to apply a 5-pixel spatial right-shift on testing datasets, assessing model resilience against geographic translation and spatial distribution shifts.
- Proved structural invariance advantages under spatial translation; the coordinate-dependent FNN suffered a complete structural collapse with accuracy plummeting from **97.46% to 17.50%**, whereas the CNN retained high residual tolerance at **56.76% accuracy**.
- Github link: Dufka54/MNIST_Spatial_Inversion: Comparative analysis between FNN and CNN. 🔗

SkyCast - Pandas | Matplotlib | Streamlit

- Engineered a resilient, keyless web application using **Streamlit** and **Pandas** to extract and visualize real-time global meteorological data.
- Coordinated a two-step data pipeline leveraging **Open-Meteo Geocoding** and **Forecast APIs** to translate natural language city queries into precise spatial coordinates (\$latitude\$/ \$longitude\$) and atmospheric metrics.
- Implemented advanced session management featuring **exponential backoff and retry logic** (urllib3) to handle network instability and rate-limiting gracefully.
- Processed complex, nested JSON payloads into clean Pandas DataFrames to generate dynamic, interactive time-series charts for multi-week temperature and precipitation trends.

Student Placement Predictor - Pandas | Scikit-Learn | Matplotlib

- Engineered a custom Linear Regression framework entirely from scratch utilizing Python and foundational mathematical principles, implementing a Batch Gradient Descent optimization algorithm to predict student placement outcomes (Binary: Placed vs. Not Placed).
- Structured a data pipeline by partitioning datasets into distinct training, validation, and testing sets. Conducted an exhaustive grid search to isolate optimal learning rates and epoch thresholds, effectively mitigating overfitting and maximizing model convergence.
- Integrated a custom classification layer by implementing a 0.5 probability decision threshold to convert continuous linear outputs into discrete binary classifications, yielding a peak predictive test accuracy of 79.25%.
- GitHub link: Dufka54/ML_firstApp 🔗 | Live Demo: mlfirstapp-2qivgrlwrddxcqc7gvmt6.streamlit.app

CERTIFICATION

Introduction to Database Systems Certificate 🔗, NPTEL 01/2025 – 04/2025
Result: 69% | Credential ID: NPTEL25CS40S647402071

Foundations of AI and Machine Learning Certificate 🔗, Microsoft 11/2025